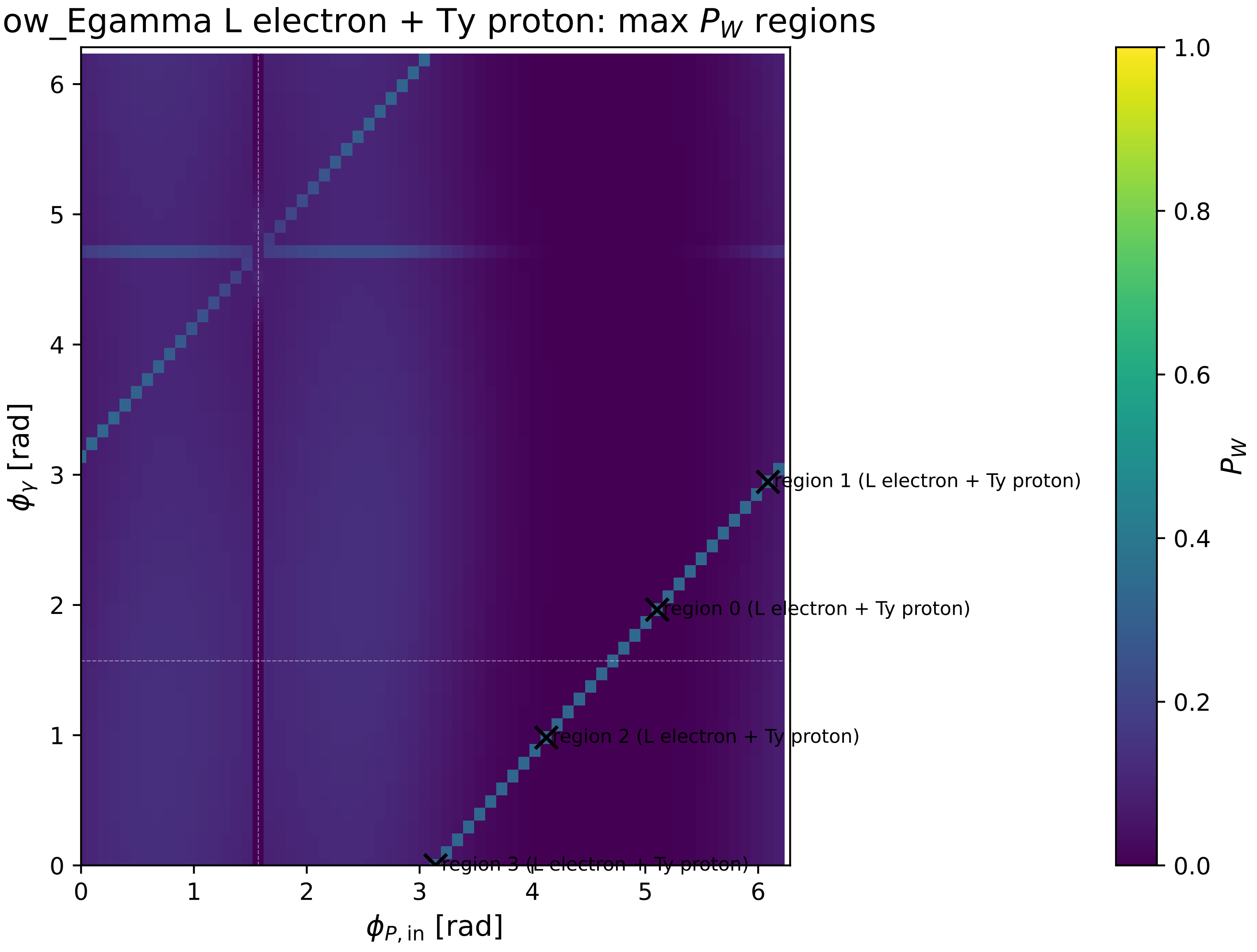
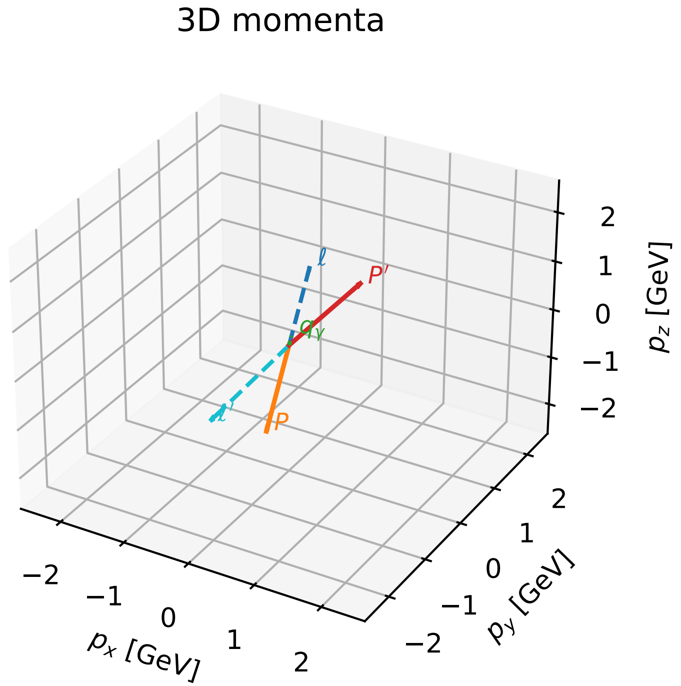

ow_Egamma L electron + Ty proton: max P_W regions
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0
region 1 (L electron + Ty proton)
region 0 (L electron + Ty proton)
region 2 (L electron + Ty proton)
region 3 (L electron + Ty proton)

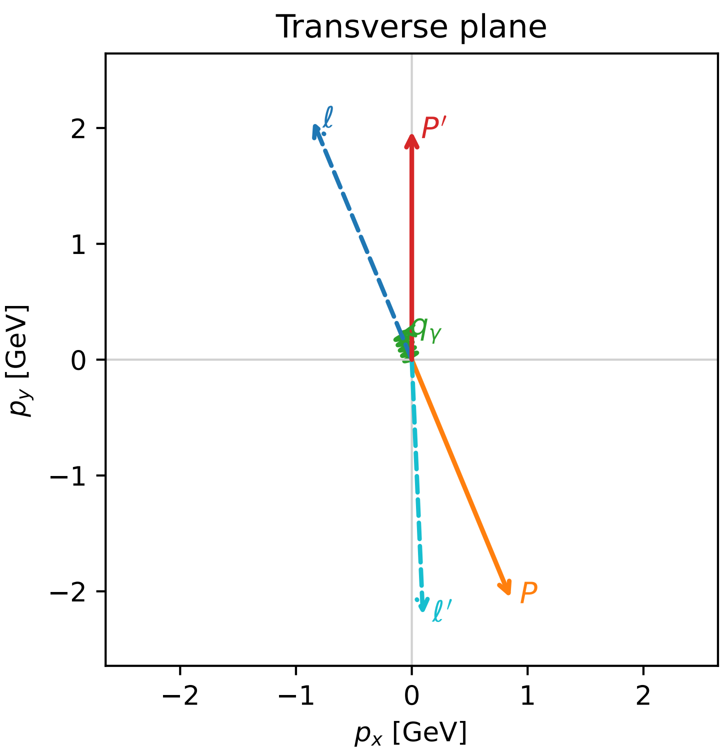
L electron + Ty proton characteristic max P_W configuration

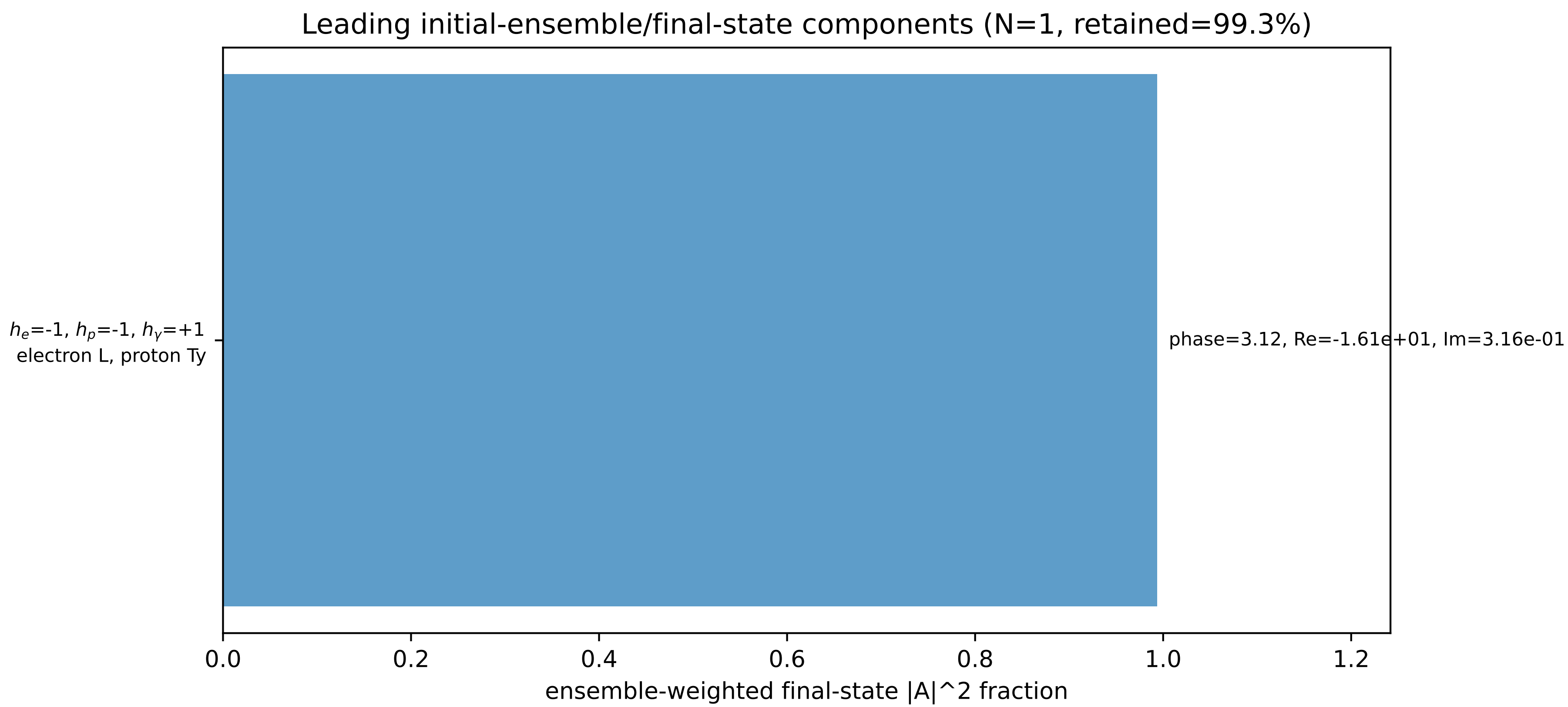


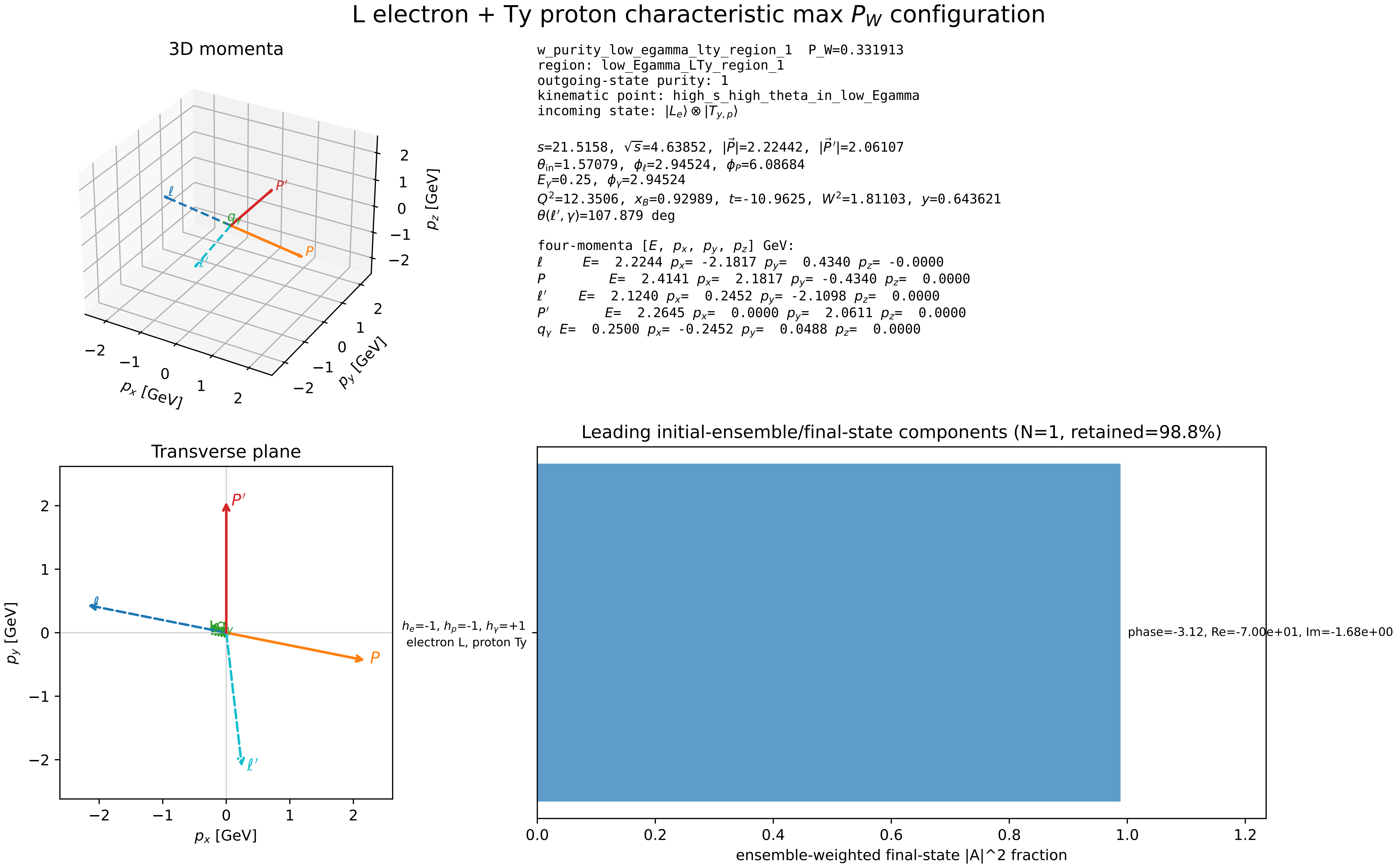
w_purity_low_egamma_lty_region_0 P_W=0.332541
region: low_Egamma_LTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.97187
θ_{in}=1.57079, ϕ_ℓ=1.9635, ϕ_P=5.10509
E_γ=0.25, ϕ_γ=1.9635
Q^2=19.0263, x_B=0.990581, t=-16.8879, W^2=1.06076, y=0.930761
$\theta(\ell', \gamma)$=159.987 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.8512 p_y= 2.0551 p_z= -0.0000
P E= 2.4141 p_x= 0.8512 p_y= -2.0551 p_z= 0.0000
ℓ' E= 2.2049 p_x= 0.0957 p_y= -2.2028 p_z= 0.0000
P' E= 2.1836 p_x= 0.0000 p_y= 1.9719 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0957 p_y= 0.2310 p_z= 0.0000

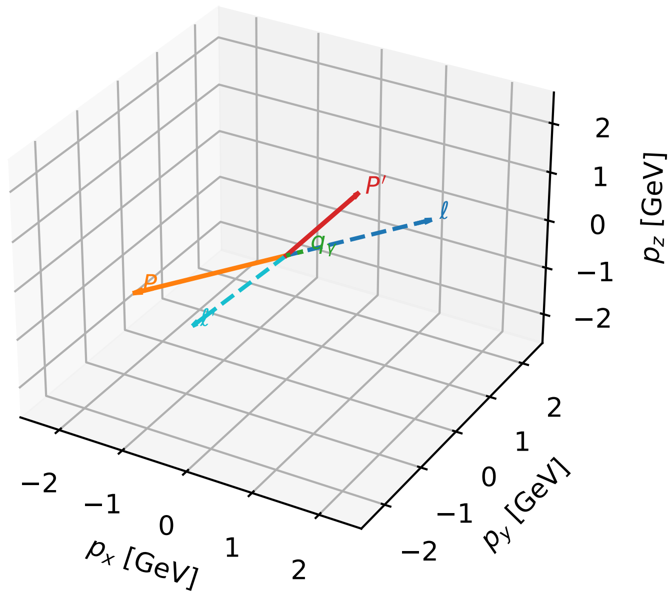





L electron + Ty proton characteristic max P_W configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ℓ
P'
q_γ
P
ℓ'
w_purity_low_egamma_lty_region_1 P_W=0.331913
region: low_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.06107
θ_{in}=1.57079, ϕ_ℓ=2.94524, ϕ_P=6.08684
E_γ=0.25, ϕ_γ=2.94524
Q^2=12.3506, x_B=0.92989, t=-10.9625, W^2=1.81103, y=0.643621
$\theta(\ell', \gamma)$=107.879 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -2.1817 p_y= 0.4340 p_z= -0.0000
P E= 2.4141 p_x= 2.1817 p_y= -0.4340 p_z= 0.0000
ℓ' E= 2.1240 p_x= 0.2452 p_y= -2.1098 p_z= 0.0000
P' E= 2.2645 p_x= 0.0000 p_y= 2.0611 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2452 p_y= 0.0488 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=1, retained=98.8%)
h_e=-1, h_p=-1, h_γ=+1
electron L, proton Ty
phase=-3.12, Re=-7.00e+01, Im=-1.68e+00
ensemble-weighted final-state |A|^2 fraction

L electron + Ty proton characteristic max P_W configuration

3D momenta

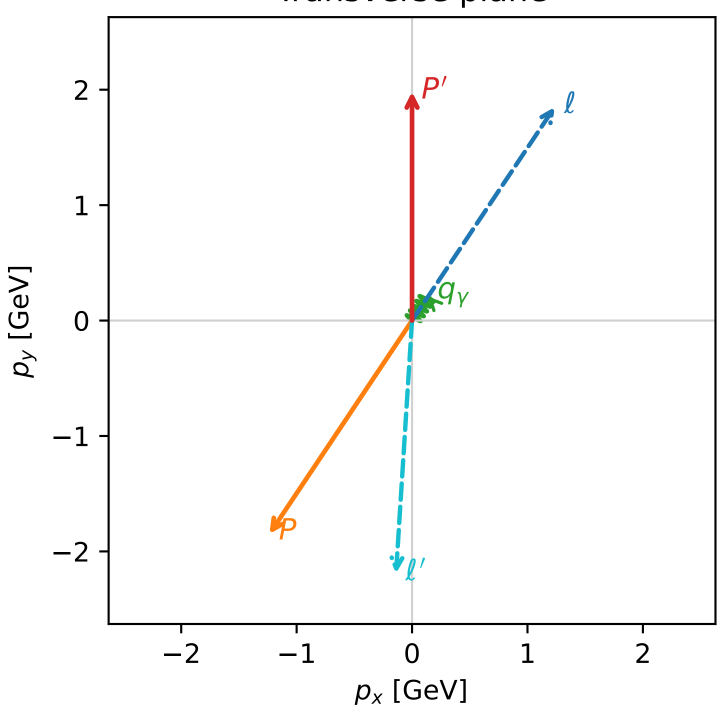



```
w_purity_low_egamma_lty_region_2  P_W=0.328682
region: low_Egamma_LTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

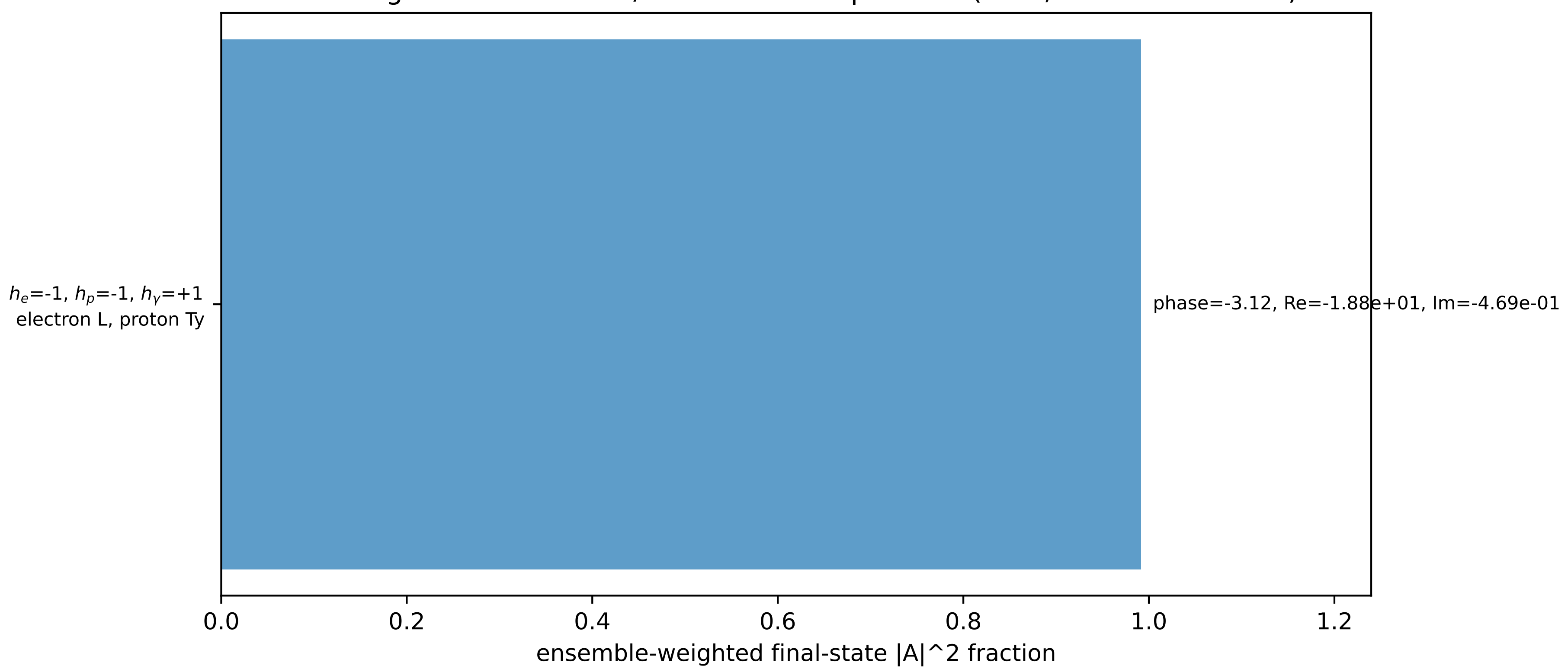
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.98279
θ_{in}=1.57079, ϕ_ℓ=0.981748, ϕ_P=4.12334
E_γ=0.25, ϕ_γ=0.981748
Q^2=18.2121, x_B=0.985262, t=-16.1653, W^2=1.15226, y=0.895742
$\theta(\ell',\gamma)$=149.878 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.2358 p_y= 1.8495 p_z= -0.0000
P E= 2.4141 p_x= -1.2358 p_y= -1.8495 p_z= 0.0000
ℓ' E= 2.1951 p_x= -0.1389 p_y= -2.1907 p_z= 0.0000
P' E= 2.1935 p_x= 0.0000 p_y= 1.9828 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1389 p_y= 0.2079 p_z= 0.0000

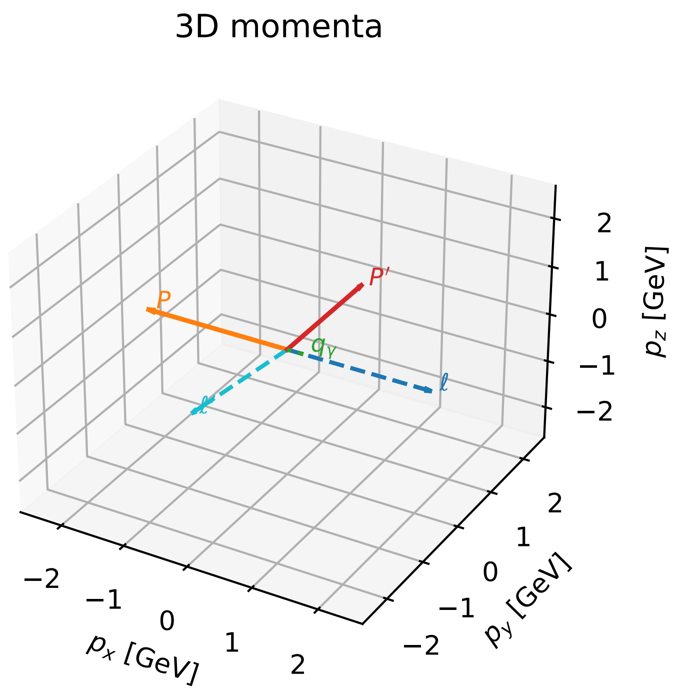
Transverse plane



Leading initial-ensemble/final-state components (N=1, retained=99.2%)



L electron + Ty proton characteristic max P_W configuration



w_purity_low_egamma_lty_region_3 P_W=0.326346
region: low_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.08621
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=0
Q^2=10.4598, x_B=0.901436, t=-9.28425, W^2=2.02353, y=0.562294
$\theta(\ell', \gamma)$=96.8334 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.1011 p_x= -0.2500 p_y= -2.0862 p_z= 0.0000
P' E= 2.2874 p_x= 0.0000 p_y= 2.0862 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2500 p_y= 0.0000 p_z= 0.0000